Lecture 4

CSE250 – Circuits and Electronics

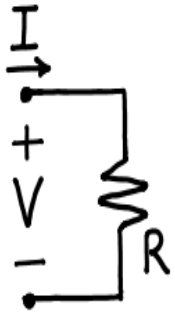
I-V of Linear Circuits

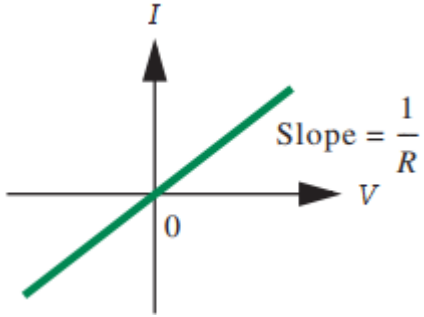


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I-V Characteristics

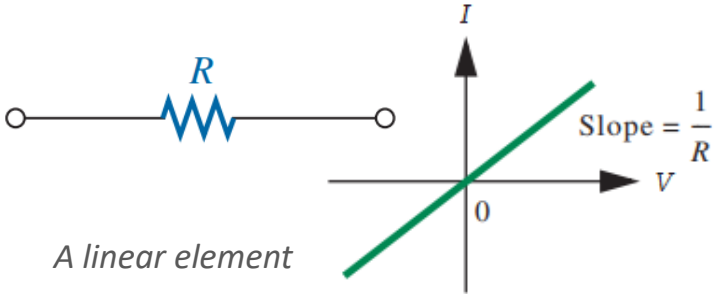
- The *current-voltage characteristics* or the $I-V$ *characteristics* is a relationship, typically expressed graphically, between the electrical current flowing through an element, circuit, device, or material and the corresponding voltage across it.
- From the viewpoint of circuit analysis, $I-V$ the most important characteristics of a two-terminal element, also called *Element Law*.
- So far, we have seen that the current voltage relationship for a resistor follows Ohm's Law, that is, $V=IR$. In an I vs. V plot it is a straight line with slope equal to $\frac{1}{R}$ that goes through the origin.
- It is important to note the direction of current to be plotted. Generally, the current plotted along the y-axis is the current drawn by the element, circuit, or device.



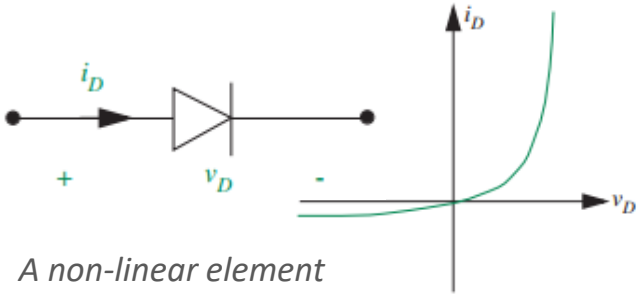


Linear vs. Non-linear Elements

- We can distinguish between the linear and non-linear circuit components from $I - V$ point of view.
- A two-terminal circuit element is said to be *linear* if it constitutes a linear (straight line) relationship between the current through and the voltage across it. Examples of linear circuit elements are ideal resistor, ideal voltage source and current source, open circuit, short circuit, capacitors, inductors etc. A circuit constructed with linear circuit elements is called a *linear circuit*.
- *Non-linear* devices, on the other hand, have a $I - V$ curve that is not straight line. Examples of non-linear elements include diodes, transistors, and nonlinear capacitors.



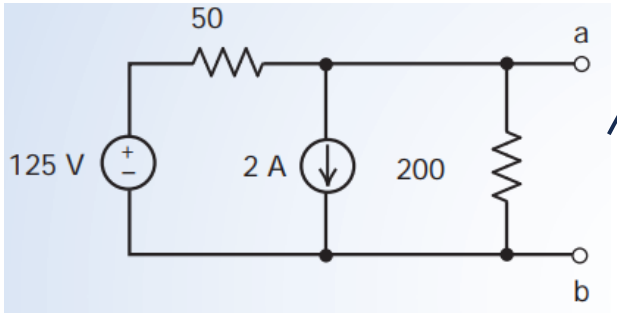
A linear element

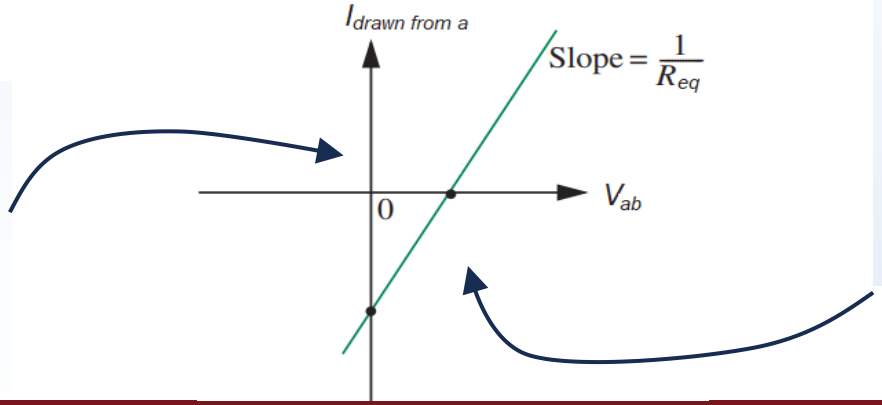


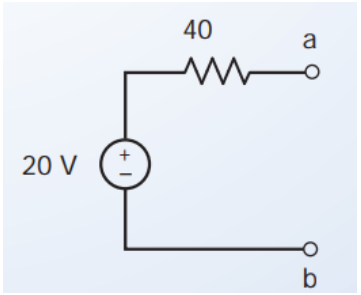
A non-linear element

Circuit Equivalence

- Two circuits are said to be linear with respect to two particular terminals (or node) if they have *identical $I - V$ relationships* between those terminals.
- For example, a resistive series network can only be replaced with their equivalent resistance if the $I - V$ line remains identical after replacement. This requires the relation $R_{eq} = R_1 + R_2 + \cdots + R_N$ to be followed.
- Similarly, the following two linear circuits are equivalent as they have the identical $I - V$ relation between terminals $a - b$ as shown. Let's see how to derive $I - V$ plots for such circuits.

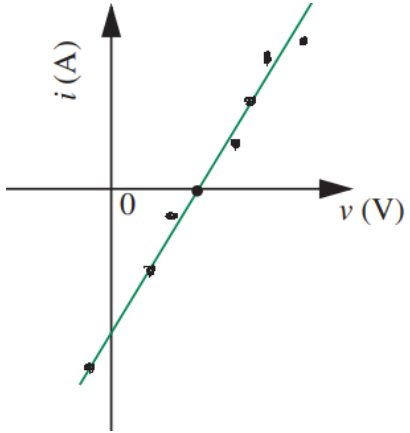






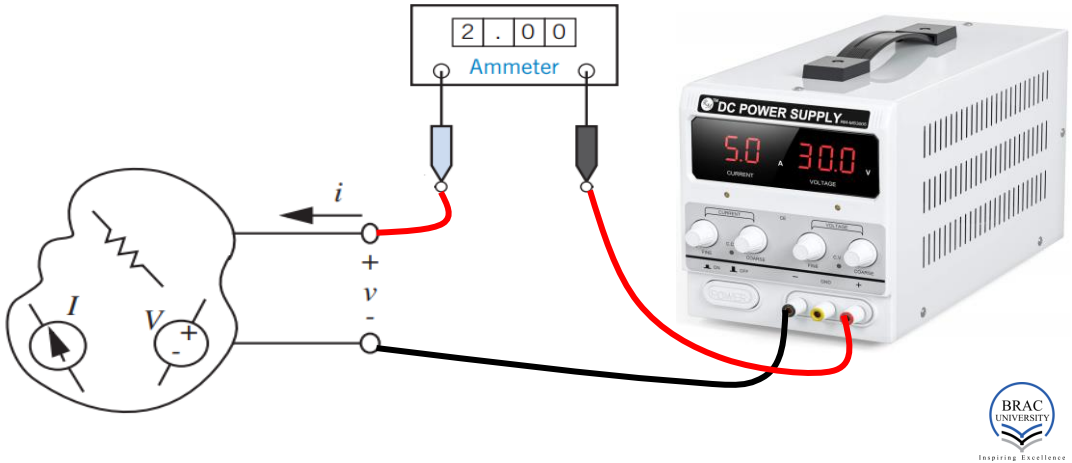
I-V: experimentally

- To determine the $I - V$ graph of a circuit experimentally, connect and vary a voltage source between the terminals where current and voltage are to be plotted.
- The varying voltages can be measured with a parallel voltmeter (or from the dc source's display), and the corresponding currents drawn by the circuitry can be measured with an ammeter in series. The more data points we collect, the more accurate the $I - V$, particularly for non-linear devices.



Data Table

v	i



I-V: theoretically

Method 1

Assume V and I variables
Assume a voltage variable between the terminals to be considered (let's say $a - b$) and a current variable directed outward from the '+' of V. The direction ensures that the current I is drawn by the circuit.

↓

Derive an equation
Apply circuit laws or other solving methods to derive a relation between the variables I and V. I and V are the only allowed variables in the equation.

↓

Plot the relation
In an I vs. V grid, plot the equation derived in step 2.

Valid only for linear circuits whose $I - V$ is a straight line.

Method 2

Apply a known voltage
Apply a voltage source between the terminals with any arbitrary value.

↓

Solve for current
Solve the circuit and determine the current supplied by the voltage source.

↓

Repeat the previous steps
Again, apply another known voltage and solve for the same current.

↓

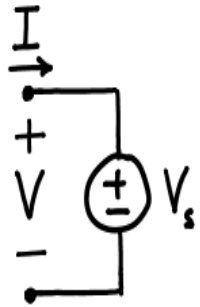
Connect the data points
Place the two data points (v_1, i_1) and (v_2, i_2) in a grid and connect them with a line.

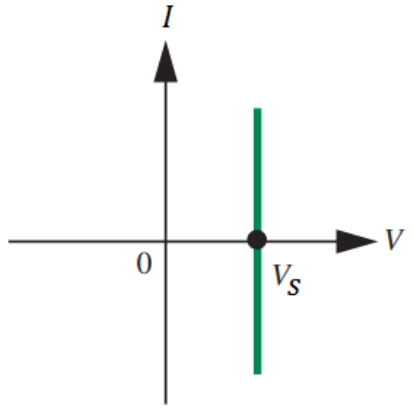
I-V of Electrical Sources

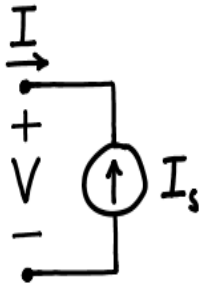
- An ideal *independent voltage source* always holds a constant potential difference between its terminals irrespective of the current drawn from it.
- So, the constituent relation for an independent voltage source supplying a voltage of V_s is,

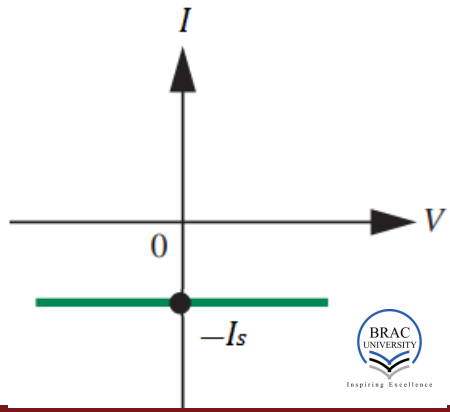
$$V = V_s$$

- This is a straight line that is parallel to the I-axis and intersects the V-axis at V_s.
- Similarly, an ideal *independent current source* always supplies a constant current to the wire it is connected irrespective of the voltage across it.
- The constituent relation is then $I = -I_s$, which is a straight line parallel to V-axis.









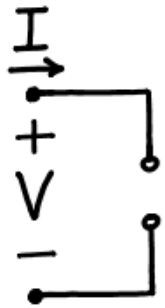
I-V of OC and SC

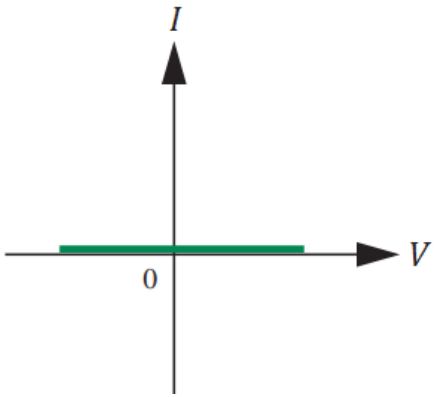
- Recall that, an *ideal open circuit* is the limiting case of a resistor where the resistance approaches infinite.
- As infinite resistance means zero current according to the Ohm's law, the constituent relation for an open circuit is,

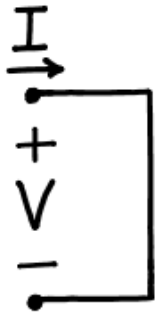
$$I = 0$$

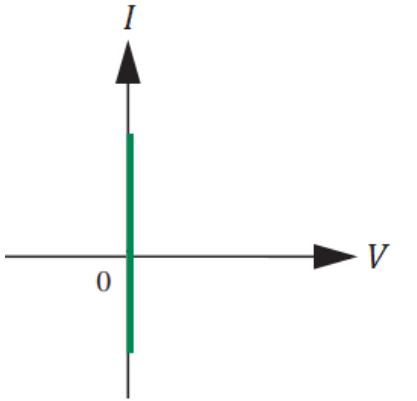
- An *ideal short circuit (or a wire)* is the limiting case of a resistor where the resistance approaches zero.
- As zero resistance means there can be no voltage difference according to the Ohm's law, the constituent relation for a short circuit is,

$$V = 0$$



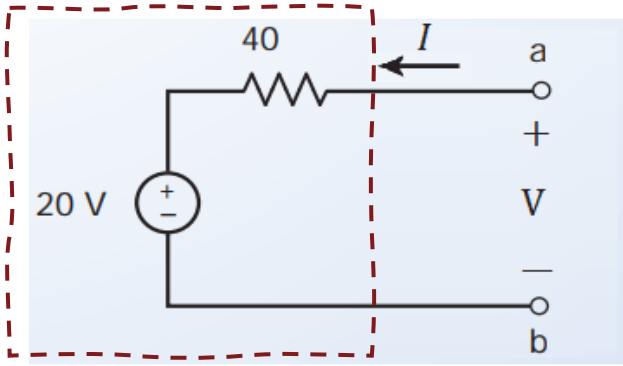






Example 1

- Derive and plot the $I - V$ relationships of the following configurations: a $20\,V$ voltage source in series with a $40\,\Omega$ resistor.

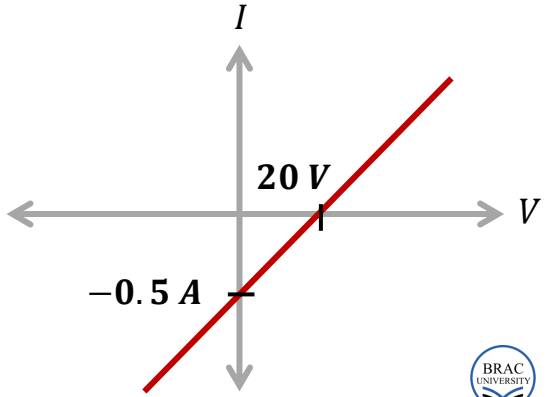


- Let's say we have a $20\,V$ voltage source in series with a $40\ \Omega$ resistor between terminals $a - b$ as shown.
- Applying KVL to the loop yields,

$$-V + 40I + 20 = 0$$

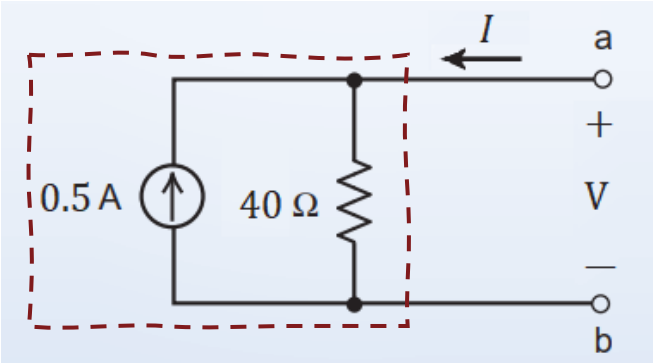
$$\Rightarrow\ I = \frac{1}{40}V - 0.5$$

- This is straight line that intersects the current and voltage axes at $(20\,V,\ 0)$ and $(0,\ -0.5\,A)$ respectively.
- *It is important to notice here that, I is the current resulting from the application of a bias V. One must not interpret the $a - b$ terminals as open circuit with 0 current in this case. Think V as a applied voltage source connected between a and b.*



Example 2

- Derive and plot the $I-V$ relationships of the following configurations: a $0.5\,A$ current source in parallel with a $40\,\Omega$ resistor.

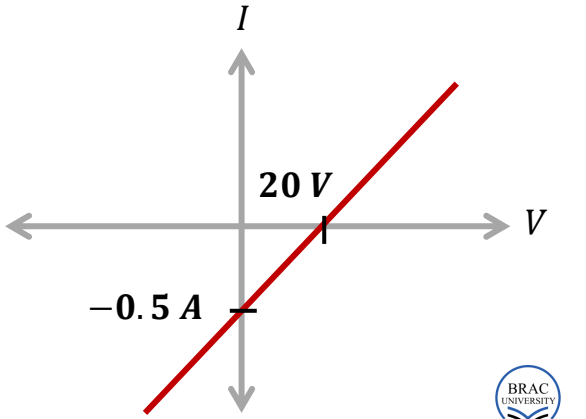


- Let's say we have a $0.5\,A$ current source in parallel with a $40\,\Omega$ resistor between terminals $a-b$ as shown.
- Applying KCL to the node a yields,

$$0.5 + I - \frac{V}{40} = 0$$

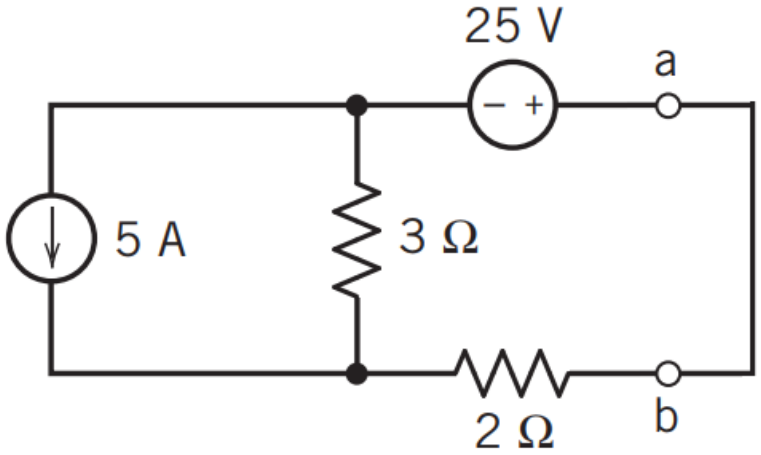
$$\Rightarrow\ I = \frac{1}{40}V - 0.5$$

- This is straight line that intersects the current and voltage axes at $(20\,V,\ 0)$ and $(0,\ -0.5\,A)$ respectively.
- *Notice that the $I-V$ curve is identical to that derived in Example 1. Thus, the two circuits are equivalent to each other.*



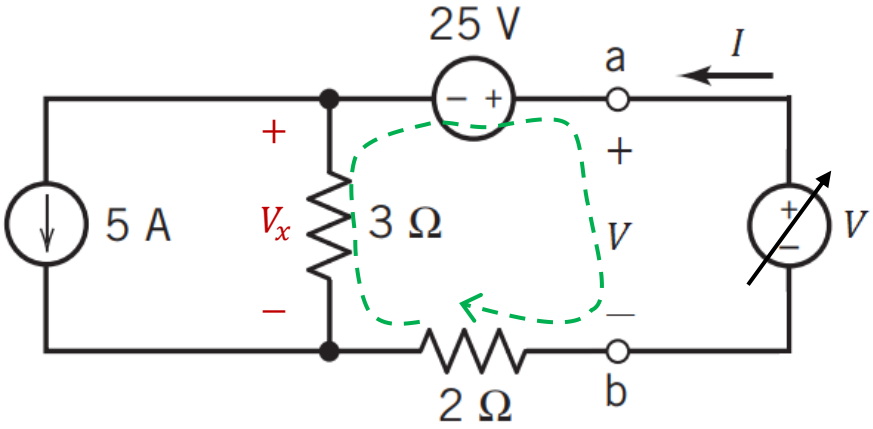
Example 3 – 1/2

- Derive and plot the $I - V$ relationship of the left portion of $a - b$ in the following circuit.



- The first step is to consider only the left portion of the terminal $a - b$. Disconnect anything connected to the right (disconnecting the short circuit in this case).
- Then we have to apply a voltage (taken as a variable V) between terminals $a - b$ and determine the current supplied by V (denoted as variable I).

Example 3 – 2/2



- The arrow symbol with a voltage source means it is a voltage to be varied, variable in our case.
- To solve the circuit using KVL, KCL, and Ohm's law, let the voltage across the $3\ \Omega$ resistor be V_x.

- Applying KCL at the positive node of V_x,

$$5 + \frac{V_x}{3} - I = 0$$

$$\Rightarrow\ V_x = 3I - 15$$

- Now, for the $25\ V$ source, we can write using KVL to the loop consisting of $3\ \Omega$, $25\ V$, $2\ \Omega$, and V as shown by the dashed green arrow,

$$-V_x - 25 + V - 2I = 0$$

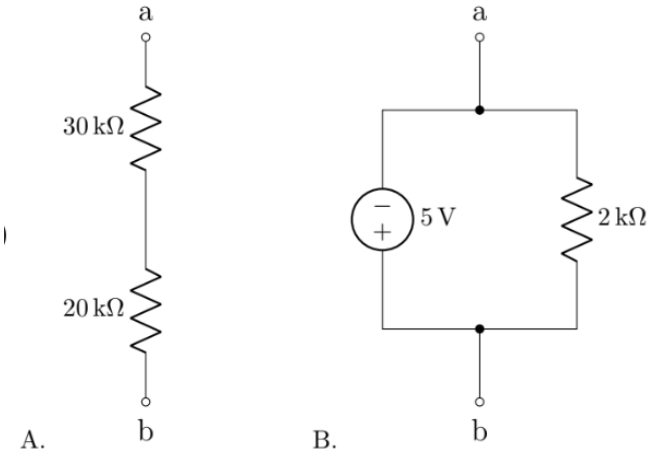
$$\Rightarrow\ -(3I - 15) - 25 + V - 2I = 0$$

$$\Rightarrow\ I = \frac{1}{5}V - 2$$

- This constitutes a straight line of slope $^1/_5\ (\Omega^{-1})$ that intersects the axes at $(10\ V,\ 0)$ and $(0,\ -2\ A)$.

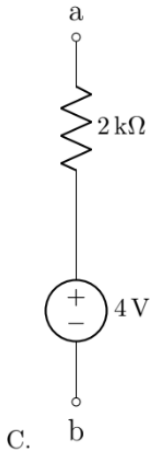
Problem 1

- Write $I - V$ characteristic equation between terminals a and b for each of the circuits shown below. Plot the I-V characteristic graphs for each of the equations.
- Are there any equivalent circuit pairs among them? Find all such sets of equivalent circuits.

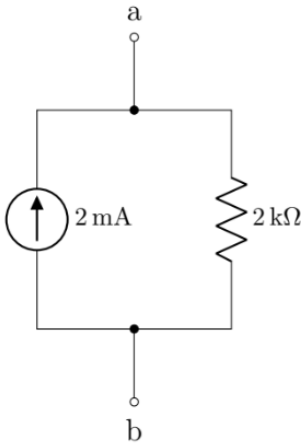


A.

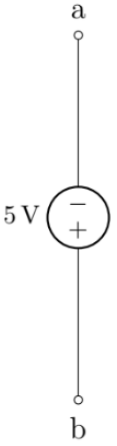
B.



C.



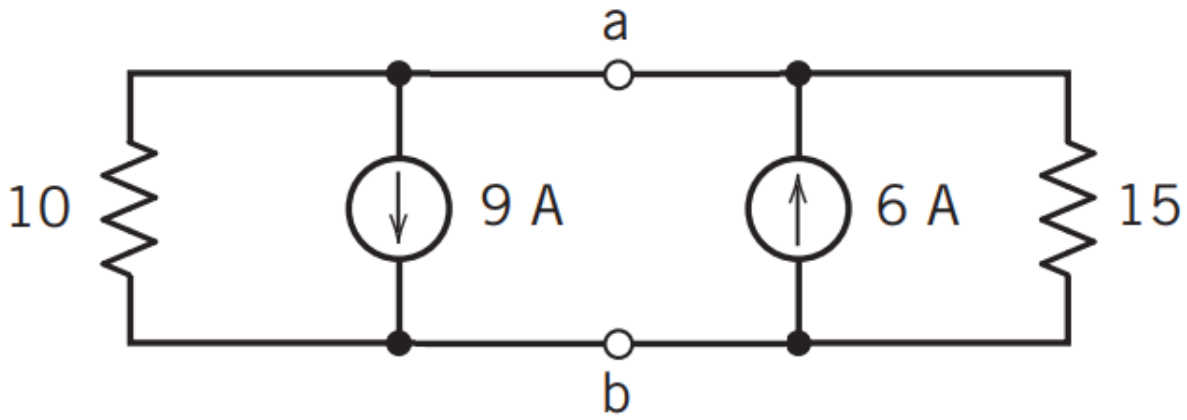
D.



E.

Problem 2

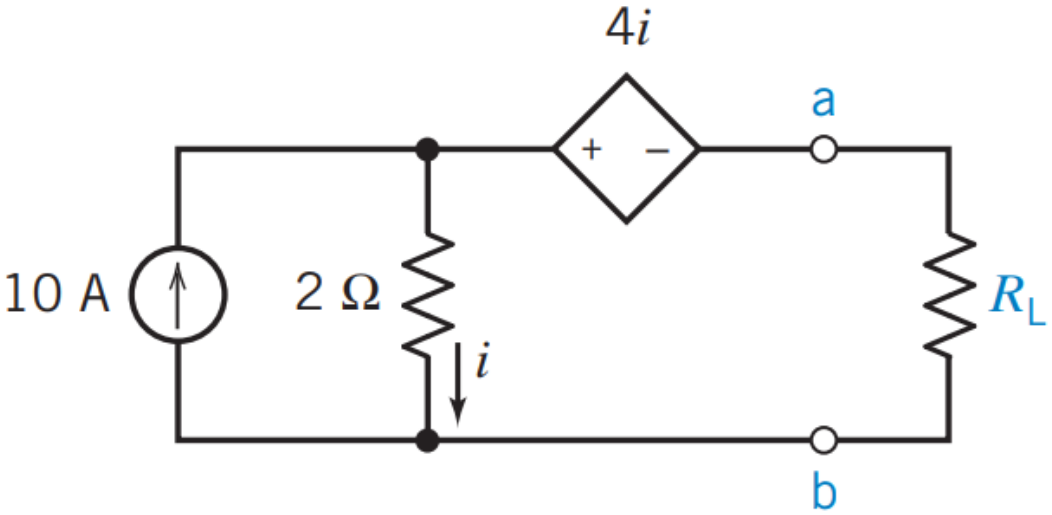
- Derive the $I - V$ characteristics of the following circuit with respect to the terminals $a - b$.



Ans: $\boldsymbol{I = \frac{1}{6}V + 3}$

Problem 3

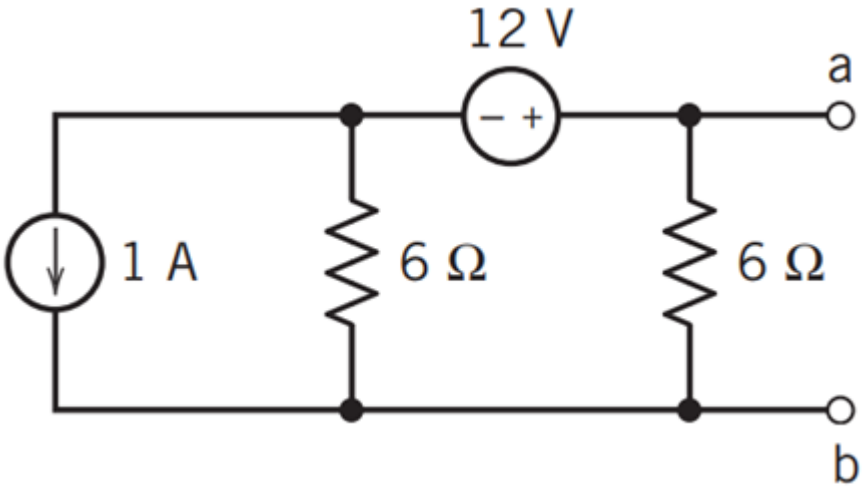
- Derive the $I - V$ characteristics of the portion left to the terminals $a - b$.



Ans: $\boldsymbol{I = -\frac{1}{2}V - 10}$

Problem 4

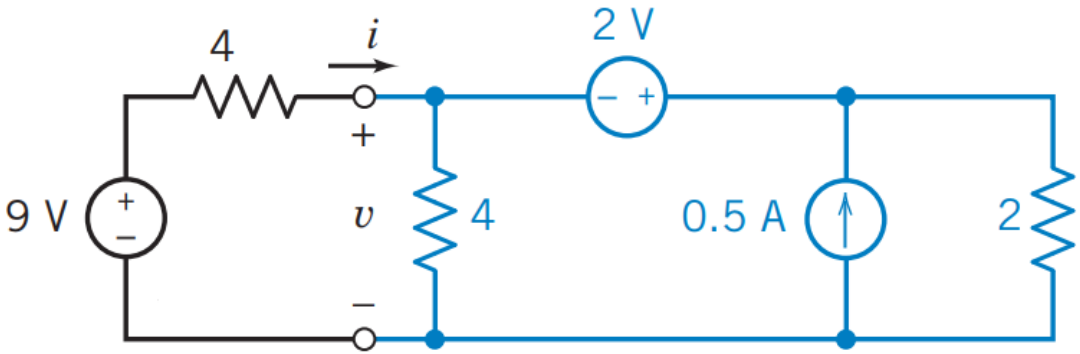
- From the following circuit, derive the current–voltage characteristics equation between the terminals $a - b$.

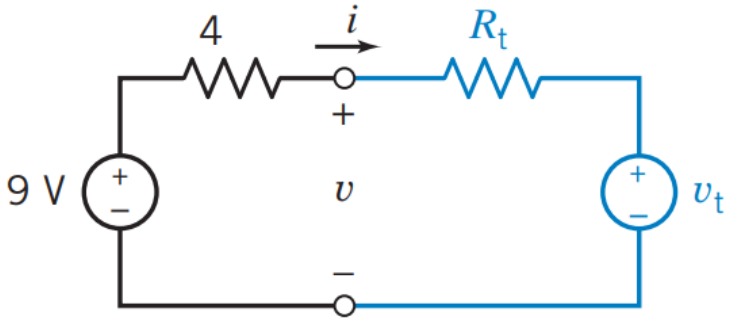


Ans: $\boldsymbol{I = \frac{1}{3}V - 1}$

Problem 5

- Determine the values of R_t and v_t, if the following two circuits are equivalent to each other.

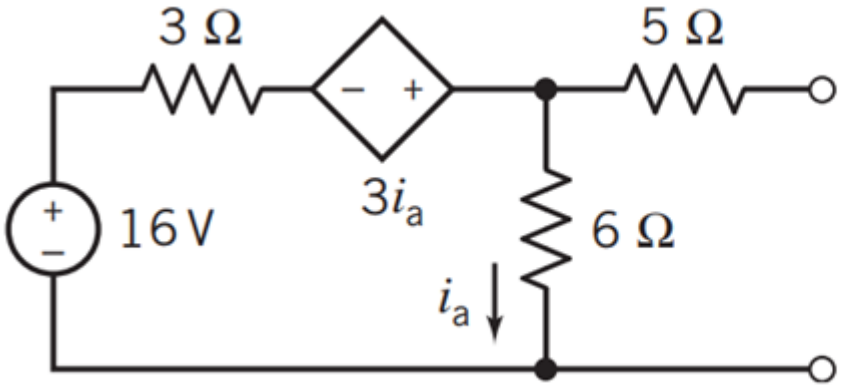


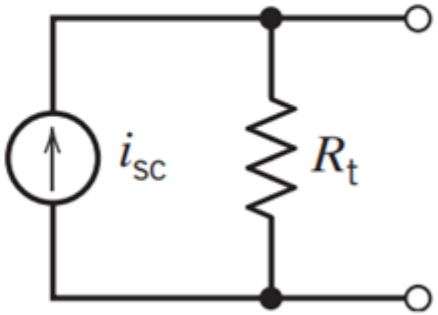


Ans: $v_t = -{}^{2}/_{3}\,V,\ R_t = {}^{4}/_{3}\ \Omega$

Problem 6

- Determine the values of R_t and i_{sc}, if the following two circuits are equivalent to each other.

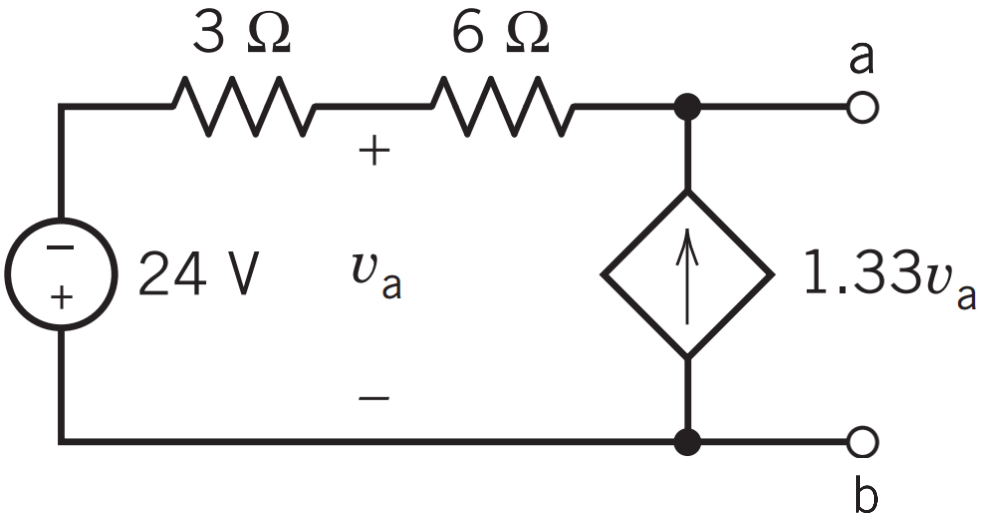




Ans: $\boldsymbol{i_{sc} = 2\ A,\ R_t = 8\ \Omega}$

Problem 7

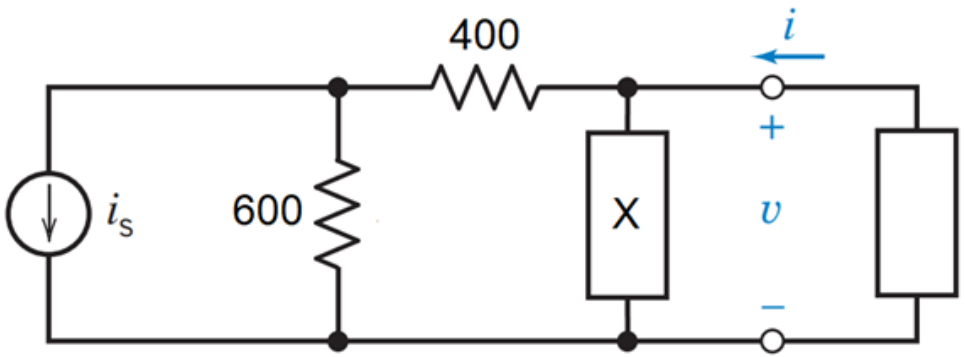
- What is the equivalent resistance of the following circuit with respect to terminals a and b?

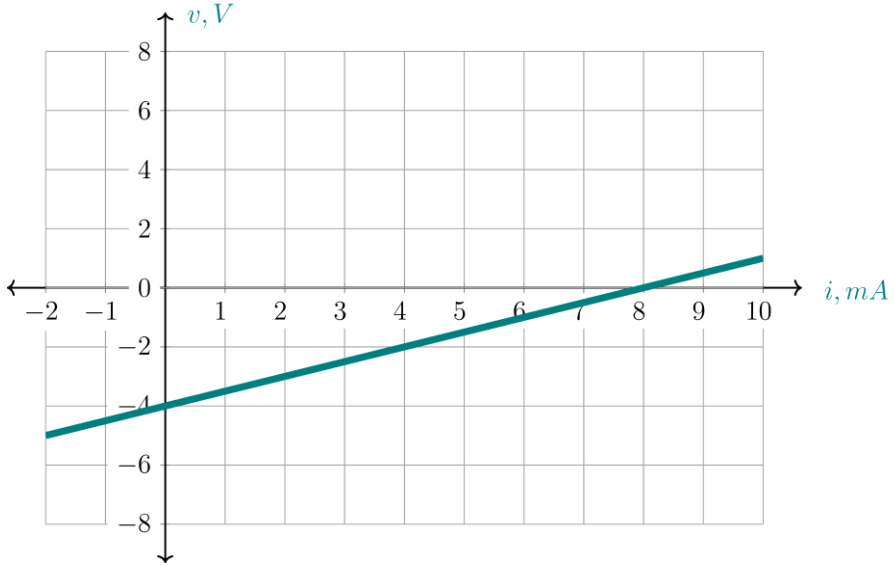


Ans: $-3\ \Omega$

Problem 8

- If the voltage v vs. current i has the following relationship expressed graphically, determine the resistance of the circuitry X.

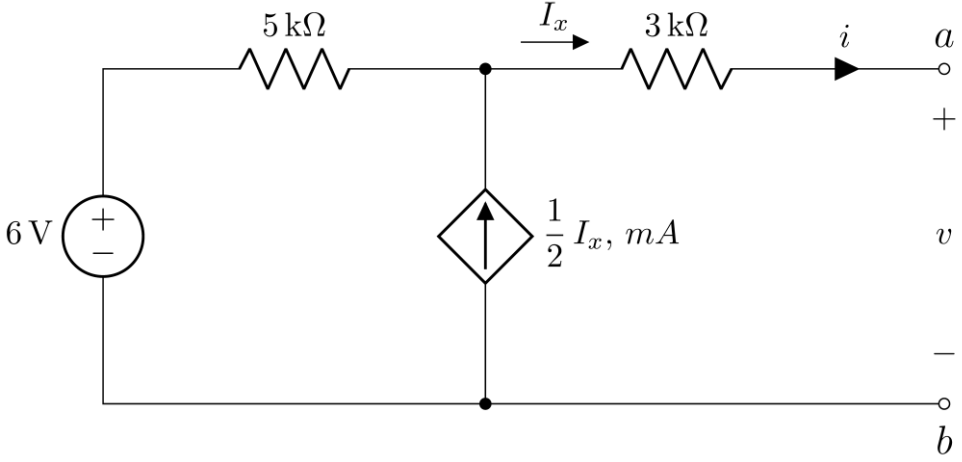




Ans: **1 $k\Omega$**

Problem 9

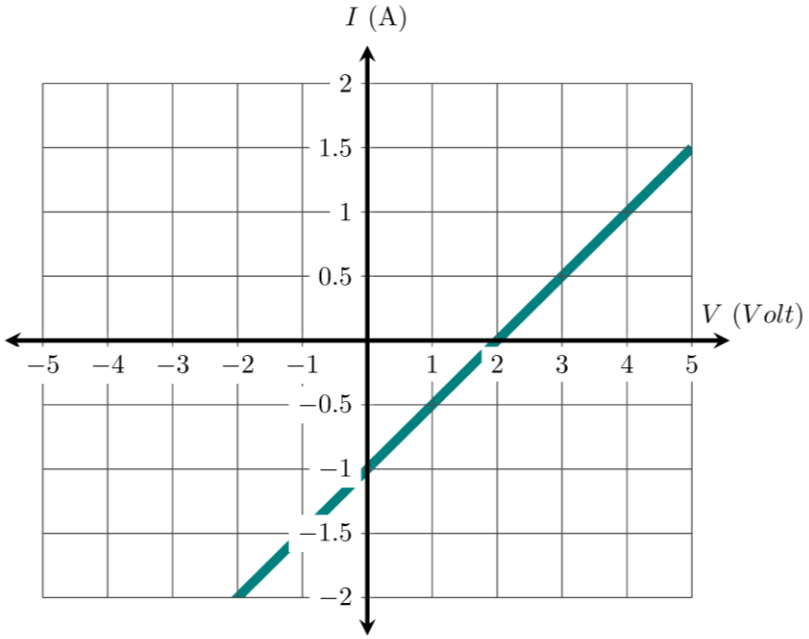
- Derive a relationship between i and v for the following circuit.

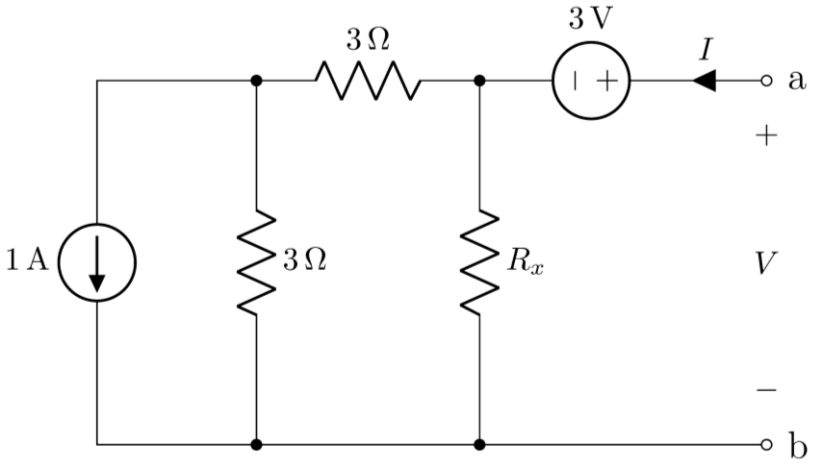


Ans: $\boldsymbol{i = -\frac{2}{11}v + \frac{12}{11}}$

Problem 10

- The circuit below has the following $I - V$ characteristic with respect to terminals a and b. Determine the unknown resistance R_x.

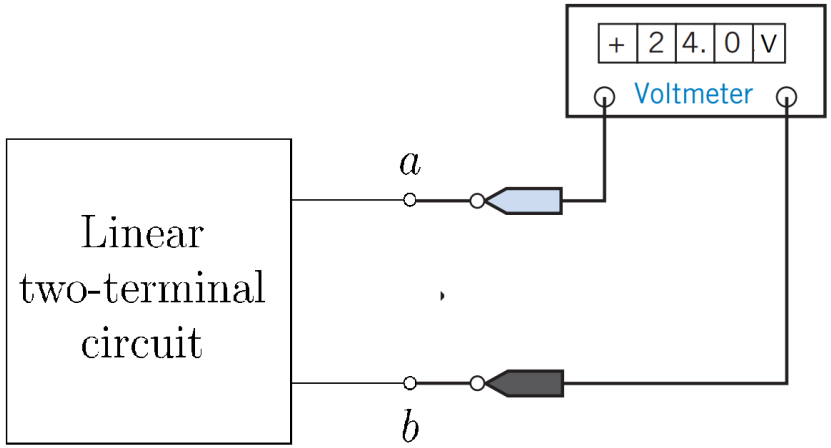


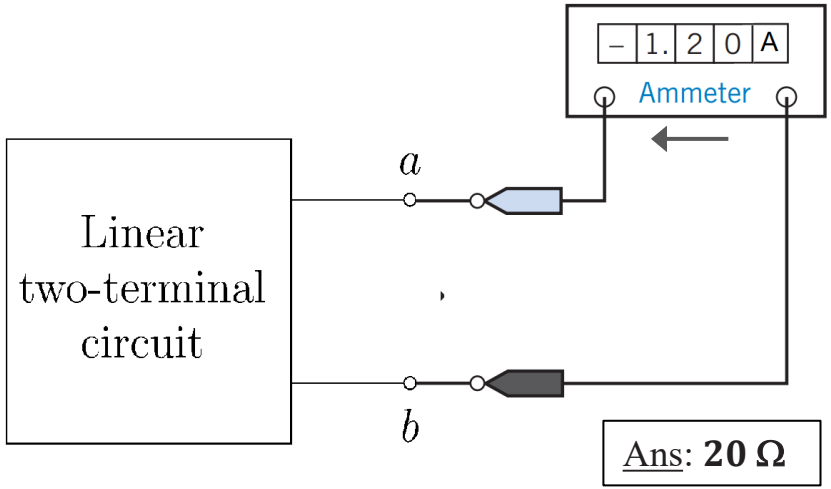


Ans: $R_x = 3\ \Omega$

Problem 11

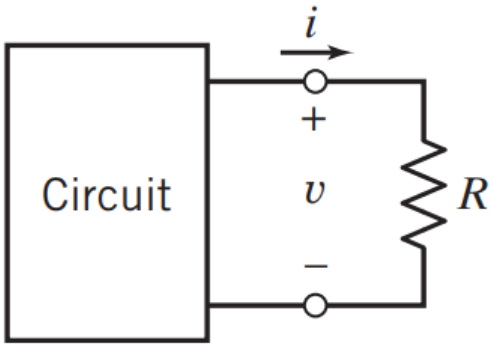
- Connecting a voltmeter and ammeter with a linear two terminal circuit shows the following measurement data. Determine the equivalent resistance of the circuit with respect to the corresponding terminals. Consider the meters ideal. *[Hint: an ideal voltmeter and an ideal ammeter have infinite and zero resistances respectively.]*





Problem 12

- A resistor, R, was connected to a circuit box as shown below. The current i was measured. The resistance was changed, and the current was measured again. The results are shown in the table.
  i. Plot the relationship between i and v.
  ii. Draw a circuit diagram with minimum number of circuit elements that can give rise to the same $i - v$ curve derived in i.

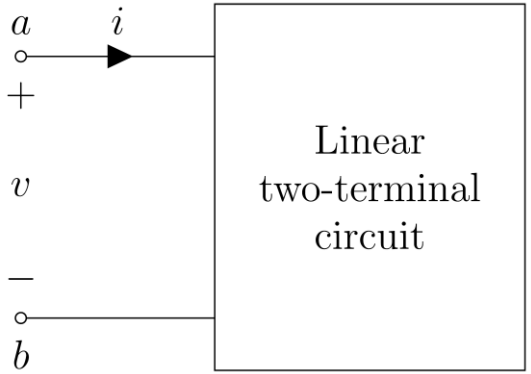


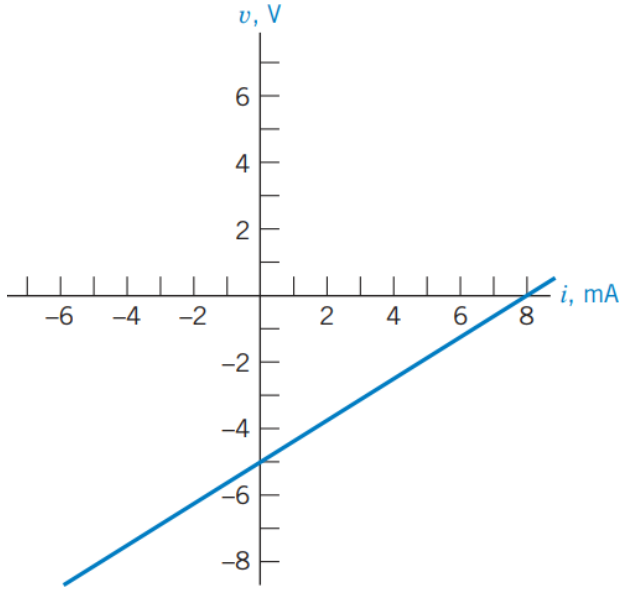
R	i
2 kΩ	4 mA
4 kΩ	3 mA

Ans: $\boldsymbol{i = -\frac{1}{4}v + 6}$

Problem 13

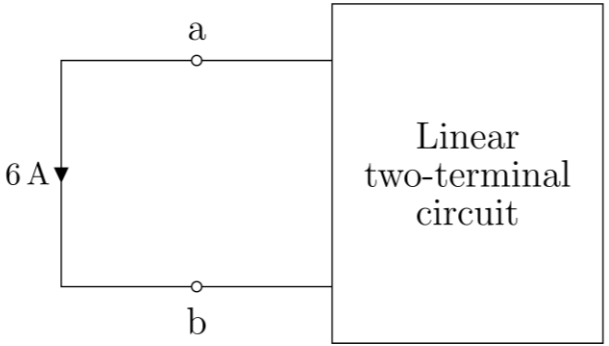
- The $V - I$ characteristic line of a linear circuit with respect to the nodes a and b are plotted below. Derive an equivalent version of the circuit with a minimum number of circuit elements so that it will give rise to the same $V - I$.

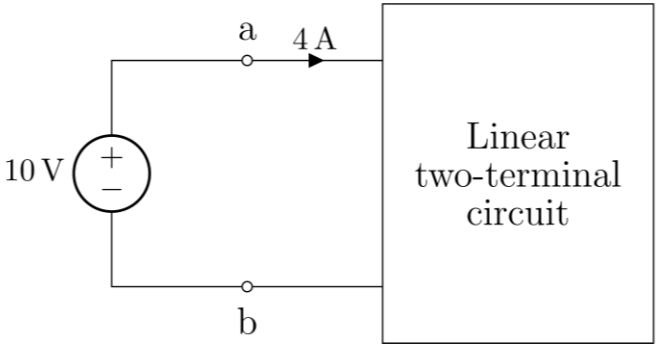




Problem 14

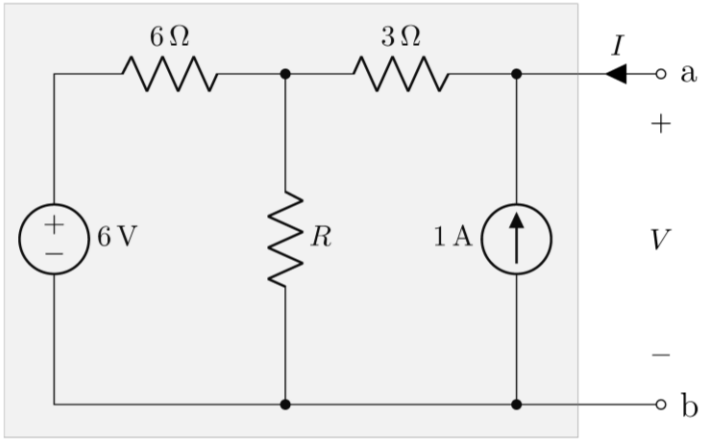
- For an unknown linear two terminal circuit as shown below, if the terminals $a - b$ are shorted, $6\ A$ current flows through the short circuit. When $10\ V$ is applied between the terminals $a - b$, the circuit draws a current equal to $4\ A$. Derive two equivalent versions of the circuit with a minimum number of circuit elements for each.

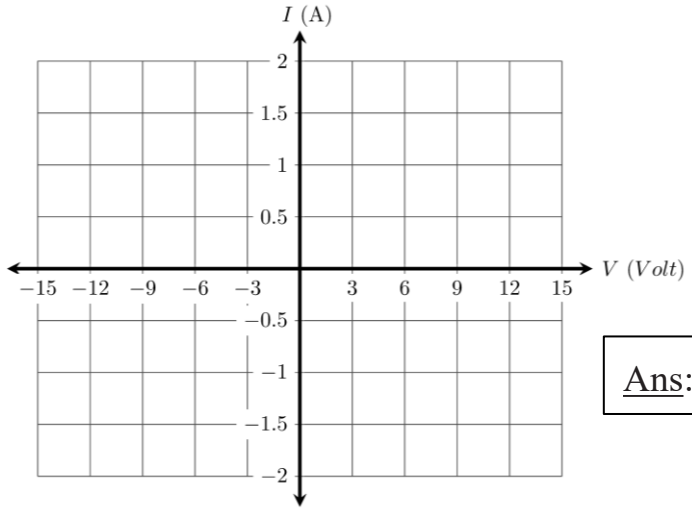




Problem 15

- If a voltage $V = 3\ V$ is applied between terminals a and b, the shaded portion of the circuit draws a current $I = -1\ A$ and when $9\ V$ is applied, it draws no current.
  a) Plot the $I - V$ characteristics of the circuit with respect to the terminals.
  b) Determine the resistance R.
  c) Draw an equivalent version of the circuit which can produce the same $I - V$ plotted in (a).

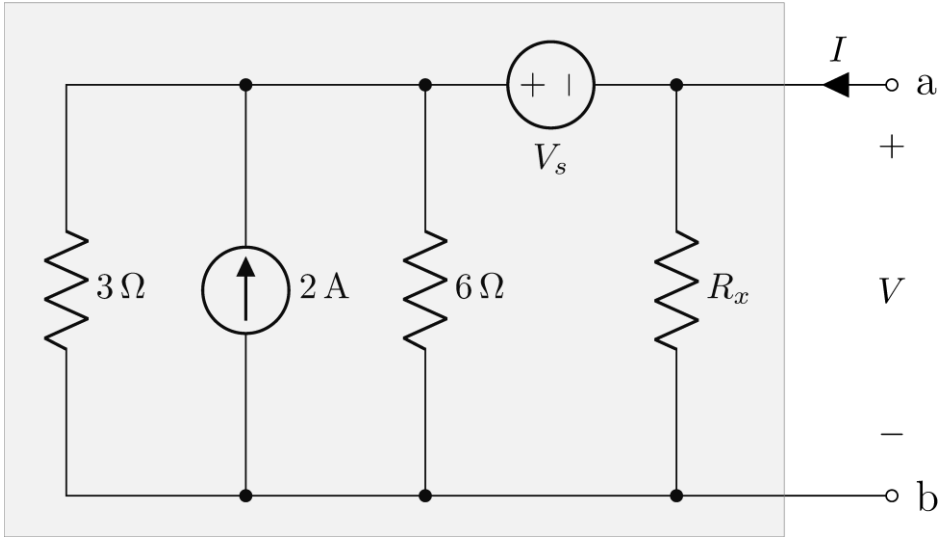




Ans: $\boldsymbol{R = 6\ \Omega}$

Problem 16

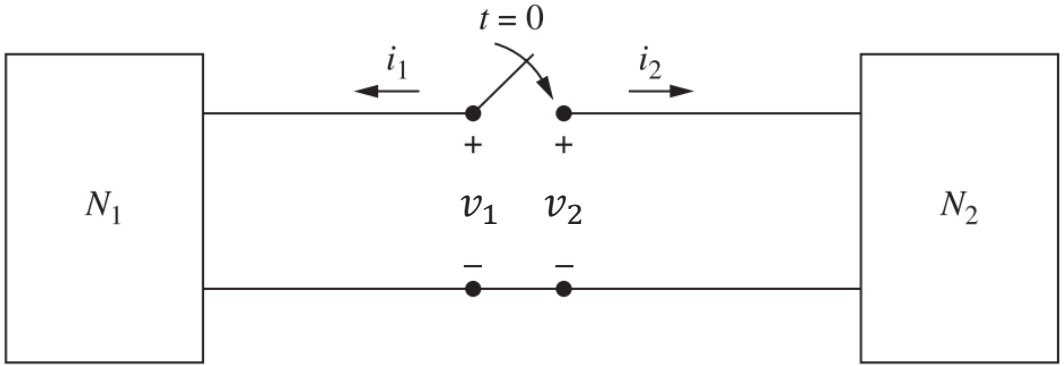
- If a voltage $V = 1\,V$ is applied between terminals a and b, the shaded portion of the circuit draws a current $I = 4\,A$ and when $0\,V$ is applied, it draws $3\,A$ current.
  a) Determine the unknown resistance R_x.
  b) Determine the unknown voltage V_s.



Ans: $\boldsymbol{R_x = 2\ \Omega, V_s = 10\ V}$

Problem 17

- Circuit N_1 is characterized by $v_1 = 2i_1 + 3$ and circuit N_2 is by $v_2 = i_2 + 6$. If the two circuits are connected at $t = 0$ by closing the switch as shown below, determine i_1, i_2, v_1, and v_2 for $t < 0$ and $t > 0$.



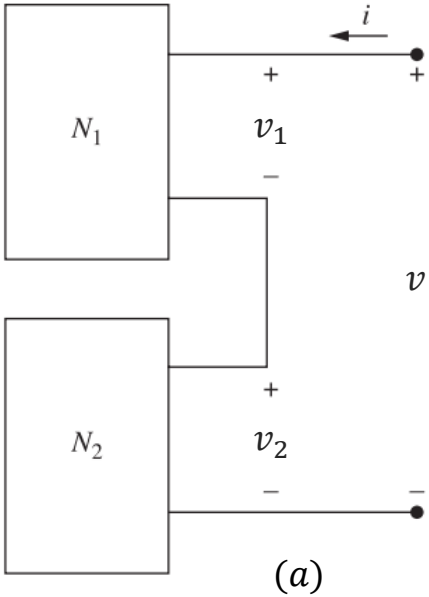
Ans:

$$\begin{cases} t < 0, \ v_1 = 3\,V, v_2 = 6\,V, \ i_1 = i_2 = 0\,A \\ t > 0 \quad\ \ v_1 = v_2 = 5\,V, \ i_1 = -i_2 = 1\,A \end{cases}$$

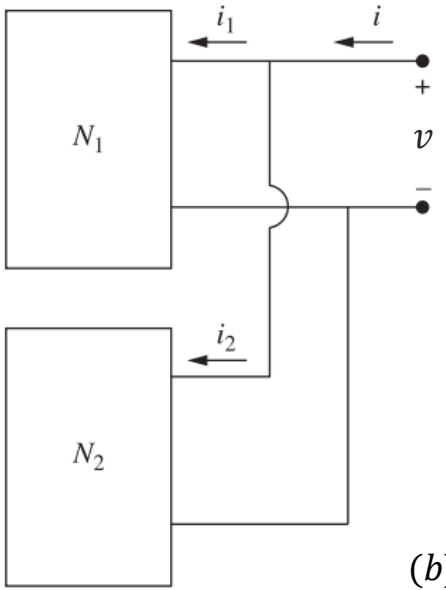
Problem 18

- Circuit N_1 is characterized by $v_1 = 2i_1 + 3$ and circuit N_2 is by $v_2 = i_2 + 6$. Find the terminal characteristic $(i - v)$ of their combination if they are connected in (a) series and (b) parallel.

Ans: $(a)\ i = \frac{1}{3}v - 3;\ (b)\ i = \frac{3}{2}v - \frac{15}{2}$



(a)

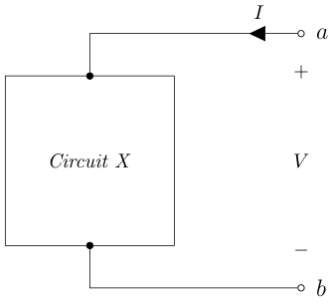


(b)

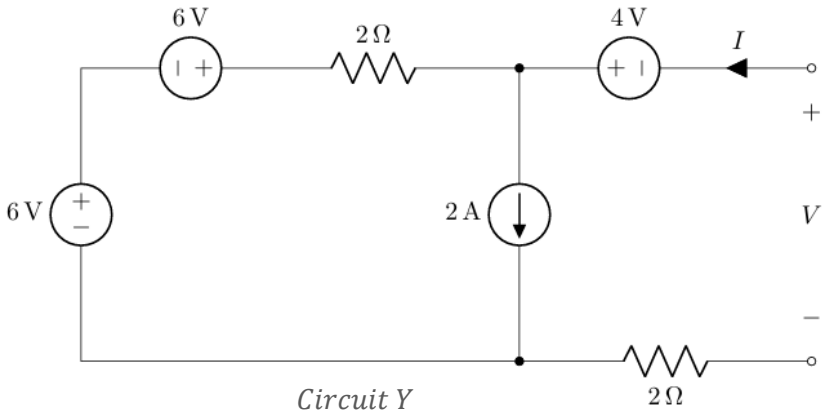
Problem 19 Summer 2025 Final

I. X is a linear two-terminal circuit connected between terminals a and b, as shown below. When tested, the circuit produced the voltage V and current I values listed in the adjacent table; however, two entries are missing. Calculate the missing values and complete the table.

II. Determine whether Circuit Y is equivalent to Circuit X in I.



V (volt)	2	6	10	
I (A)	−0.5	0.5		2.0



Ans: ***I.*** $(10\ \mathbf{V},\ 1.5\ \mathbf{A}), (12\ \mathbf{V}, 2.0\ \mathbf{A})$; ***II. Equivalent***

Problem 20 Summer 2024 Final

I. Derive a relation between I and V for Circuit X.

II. If the circuit in *Figure 3* is an alternative version of Circuit X, determine V' and R'.

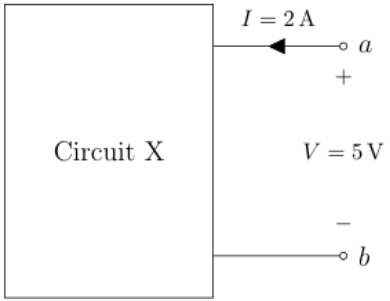


Figure 1

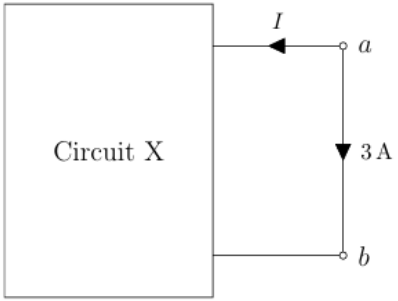


Figure 2

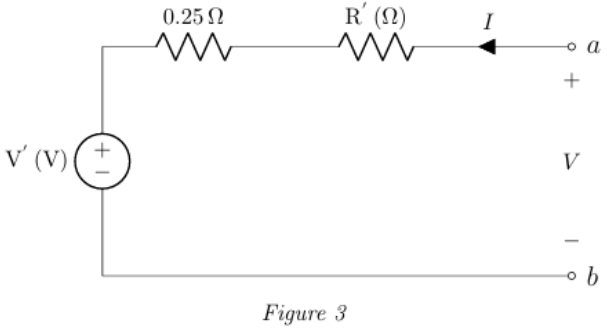


Figure 3

Ans: $\boldsymbol{I.\ I = V - 3;\ II.\ V' = 3\,V,\ R' = 0.75\,\Omega}$

Acknowledgements and References

- Some of the problems, illustrations, and concepts in this lecture are taken from the following sources:

1. Agarwal, A., & Lang, J., Foundations of Analog and Digital Electronic Circuits, Morgan Kaufmann
2. Dorf, R. C., & Svoboda, J. A., Introduction to Electric Circuits, Wiley
3. Nahvi, M., & Edminister, J., Schaum's Outline of Electric Circuits, McGraw-Hill

Thank you for your attention